Team 34 / Year 2026-2027 / Project “Rock Your Baby”

# Complete Design Document

Version 1.5 (Sections 1+2)

15/10/2026

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# 1 Requirements and Architecture

## 1.1 Architecture and Requirements at Product Level (System level)

**1.1.1 Goal of the product** The product is an electronic controller for a baby cradle that settles a distressed baby without anyone having to keep rocking it by hand. It senses how stressed the baby is from two signals only: how loudly the baby cries and how fast its heartbeat runs. It then chooses the rocking frequency and amplitude of the cradle so that the baby's stress drops to a calm level as quickly as possible, relieving caregivers of constant manual soothing and making the settling repeatable rather than a matter of guesswork.

| Description 1.1.1 | Max Points |
|---|---|
| - The goal of the product is clearly explained - 2 points<br>- Anything else - 0 point. | 2 |

**1.1.2 Set of Requirements at Product level (System Level)** **PR-1.** The product shall reduce the baby's stress level from its initial distressed state to the relaxed state without any manual interference from outside (goal of the challenge [1], ch. 1).

**PR-2.** The product shall determine the baby's stress level continuously, based on exactly two input signals: the crying intensity (sound) and the heartbeat rate (flashing LED).

**PR-3.** The product shall output two independent commands, rocking frequency and rocking amplitude, based on the external variables, and keep the commanded rocking motion constant long enough for the baby model to react.

**PR-4.** The product shall detect when the current rocking motion is wrong (panic block or panic jump behavior in the inverse model [1]) and switch to a safe neighbouring motion, instead of sticking to a failing motion.

**PR-5.** The product shall communicate all measured and decided values between the four sub-module PYNQ boards, and show them on the LCD display of each sub-module (display requirement, ch. 2.5).

| Description 1.1.2 | Max Points |
|---|---|
| - At least 1 valid requirement relating to the need of each of the 5 submodules is given – 2 point.<br>- Anything else - 0 point. | 2 |

**1.1.3 Description of the Inputs and Outputs at Product level (System Level)** The product has two inputs and two outputs.

**Input 1 — Crying (sound)**

The baby doll cries through a speaker built into the cradle, next to the head of the doll. A fixed dynamic microphone, already mounted on the cradle [1], picks up the sound. For stress levels between 50% and 100% the crying volume is constant; below 50% stress the volume drops linearly with the stress. The loudness of the cry is therefore treated as a direct, monotonic measure of the lower half of the stress range.

**Input 2 — Heartbeat (light)**

The heartbeat of the baby is a flashing LED under the skin of the baby's arm, in the wrist. Its rate runs from 60 BPM (relaxed) to 240 BPM (panic), rising with stress after a fixed delay. The LED can emit four emulations: a clean 50% square wave, a square wave with glitches, a square wave with rate deviations, and an ECG pattern. The sensor must handle all four, because the grader may select any of them (Setup Start-up Guide, settings).

**Output 1 — Rocking frequency**

A 1 kHz, 12 V PWM signal (PWM_F) whose duty cycle selects one of five rocking frequencies: 0.2, 0.35, 0.5, 0.65, 0.70 Hz. The duty windows per step are fixed by the challenge (F1: 0-10%, F2: 10-30%, F3: 30-50%, F4: 50-70%, F5: 70-90%).

**Output 2 — Rocking amplitude**

A 1 kHz, 12 V PWM signal (PWM_A) whose duty cycle selects one of five rocking amplitudes: 20%, 40%, 60%, 80%, 100%. Same duty windows as for frequency (A1-A5).

Both PWM signals must supply at least 0.8 A to the swing motors and must never exceed 90% duty.

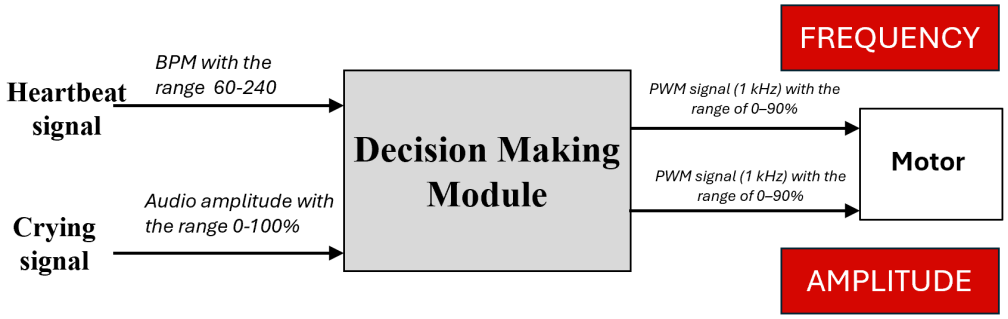



Figure 1: High-level context diagram of the inputs and outputs of the product.

| Description 1.1.3 | Max Points |
|---|---|
| - The 2 inputs and 2 outputs of the system are well described and identified using extra non-text material – 3 point.<br>- The 2 inputs and 2 outputs of the system are given but their description and/or identification are lacking – 1,5 point.<br>- Anything else - 0 point. | 3 |

**1.1.4 Specification of the Inputs and Outputs at Product level (System Level)** Signal specifications, extracted from the challenge document. These four are the product boundary; internal signals are named and justified in 1.3 and specified per sub-module in Section 3.

**CRY_RAW — crying sound (input, to S2)**

| Name | CRY_RAW |
|---|---|
| Description | Acoustic crying signal of the baby, picked up by the fixed dynamic microphone on the cradle. |
| Unit | volt (V) |
| Data type | analog AC voltage |
| Range | 1 V peak-to-peak or more at full crying volume after amplification (Demo-1 threshold). Below 50% stress, the p-p value scales down linearly with stress |
| Timing / sampling | voice content ~50 Hz to 8 kHz; sampling at $\geq$ 16 kHz, cry level recomputed at a 10 Hz refresh |
| Notes | The volume is constant while stress is in (50, 100]. Therefore, the crying sound is only usable to resolve stress below 50% |

**HB_RAW — heartbeat optical signal (input, to S1)**

| Name | HB_RAW |
|---|---|
| Description | Light pulses from the heartbeat LED in the baby's wrist. Intensity and shape follow one of four emulations: 50% square; square + glitches; square with rate deviations; ECG. |
| Unit | photons → converted to volt at the sensor. |
| Data type | analog optical pulse, digitized to a 0/3.3 V square at the S1 output |
| Range | 60 to 240 BPM (1 to 4 Hz fundamental pulse rate) |
| Timing / sampling | at 240 BPM the pulse period is 250 ms; the conditioned signal is sampled at $\geq$ 40 samples per period ($\geq$ 160 Hz), the BPM value is updated at 2 Hz |
| Notes | The rate lags the true stress by a fixed, unknown delay of the model. The algorithm (S3) must tolerate it. |

**PWM_F — rocking frequency command (output, from S4)**

| Name | PWM_F |
|---|---|
| Description | PWM command to the cradle swing motor that sets the rocking frequency. |
| Unit | duty cycle in percent (0-90) |
| Data type | digital PWM, 1 kHz carrier, 12 V amplitude |
| Range | F1 0-10% / F2 10-30% / F3 30-50% / F4 50-70% / F5 70-90% |
| Timing / sampling | 1 kHz carrier; command value may change at most every 1 s (see PR-3: give the model time to converge) |
| Notes | $\geq$ 0.8 A available to the motor; duty > 90% trips the red emergency light and is forbidden |

**PWM_A — rocking amplitude command (output, from S4)**

| Name | PWM_A |
|---|---|
| Description | PWM command to the cradle swing motor that sets the rocking amplitude. |
| Unit | duty cycle in percent (0-90) |
| Data type | digital PWM, 1 kHz carrier, 12 V amplitude |
| Range | A1 0-10% / A2 10-30% / A3 30-50% / A4 50-70% / A5 70-90% → rocking at 20% / 40% / 60% / 80% / 100% amplitude |
| Timing / sampling | 1 kHz carrier; same 1 s minimum hold time as PWM_F |
| Notes | $\geq$ 0.8 A; duty > 90% forbidden |

| Description 1.1.4 | Max Points |
|---|---|
| - The 2 inputs and 2 outputs of the system are fully specified(unit, data type, range, timing information)– 3 point.<br>- 1 (out of the 4) signal specification is not fully specified – 1,5 point.<br>- Anything else - 0 point. | 3 |

## 1.2 Architecture and Requirements at Submodule Level (Sub-system level)

**1.2.1 Overview of the product with a high-level context diagram** The product consists of four sub-modules, each implemented on its own PYNQ-Z2 board with an LCD display, all four boards mounted in the cradle backbone [2]. The sub-modules are:

**S1 Heartbeat** — detect the flashing light, digitize it, compute the heartbeat BPM, and send the variable to the decision making module.

**S2 Crying** — correctly interpret the analog microphone signal and filter the noise (signal conditioning), convert it into a digital signal with a range from 0-100%,and communicate it to the decision making module.

**S3 Decision Making** — estimates the stress from the two measurements and decides the rocking motion (frequency/amplitude).

**S4 Motor drivers** — converts the (F, A) decision into the two 1 kHz PWM signals driving the cradle swing motors.

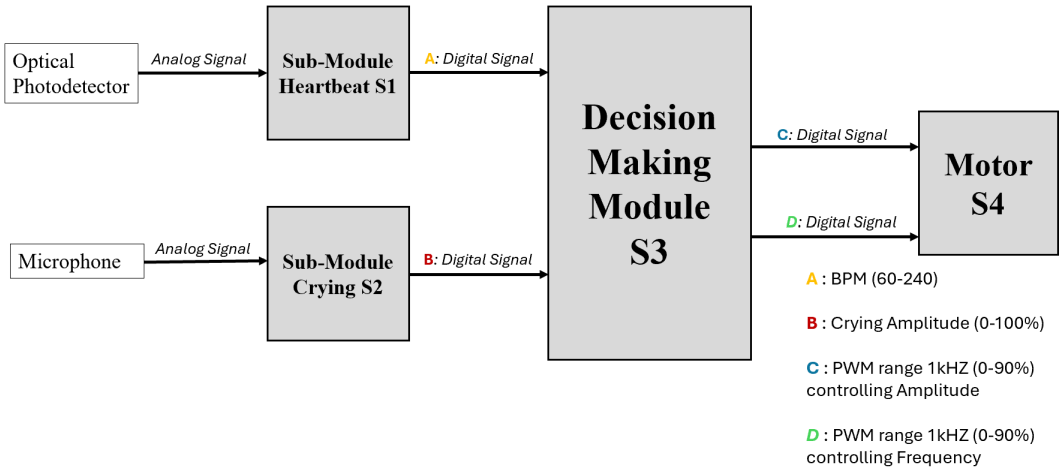



Figure 2: Overview of the different submodules.

| Description 1.2.1 | Max Points |
|---|---|
| - A brief (and correct) overview of the product and its submodules AND a high-level context diagram is given – 2.5 point.<br>- The overview OR the diagram is missing/incorrect – 1 point.<br>- Anything else - 0 point. | 3 |

### 1.2.2 Goal of each submodule

**Goal of Submodule "Heartbeat" (S1)** Sense the flashing heartbeat LED around the baby's wrist. The output must be a BPM value within ±5 BPM of the true rate, plus its short-term trend, and be communicated to the decision making module. It must work for all four LED emulations over the full 60-240 BPM range.

**Goal of Submodule "Crying" (S2)** Measure how loudly the baby is crying from the fixed microphone, and output a normalized cry-level value (0-100%) plus its trend. The output must track the baby's crying volume, and communicate it to the decision making module.

**Goal of Submodule "Decision Making" (S3)** Fuse the heartbeat and crying measurements into an estimate of the baby's stress level, and choose the next rocking motion (F, A).

**Goal of Submodule "Motors drivers" (S4)** Convert the chosen motion (F, A) into two PWM signals, clamped to 90% duty. It must settle any change within 200 ms but never let the duty exceed the emergency limit.

| **Description 1.2.2** | **Max Points** |
|---|---|
| Distribution of the score : 0.5 point per correct concise description of the Goal of each Submodule.<br>- Heartbeat<br>- Crying<br>- Decision making<br>- Motor drivers | 2 |

### 1.2.3 Set of requirements for each submodule

**Set of requirements for the Submodule "Heartbeat"** **SR-HB-1**: S1 shall output a BPM value in the range 60-240 BPM with a refresh rate of at least 2 Hz.

**SR-HB-2**: S1 shall measure the true BPM within ±5 BPM for all four LED emulations (square 50%, square with glitches, square with rate deviations, ECG pattern).

**SR-HB-3**: S1 shall attach a trend flag (rising / stable / falling) to the BPM value, computed over at least 3 consecutive updates.

**SR-HB-4**: S1 shall mount on the baby's wrist without adhesive tape (Velcro, cable ties or rubber bands only) and shall not modify the baby setup.

**SR-HB-5** (display): S1's LCD shall show the current BPM, the trend flag, and the raw waveform status, refreshed at 2 Hz.

**Set of requirements for the Submodule "Crying"** **SR-CR-1**: S2 shall output a normalized cry level in 0-100% with a refresh rate of at least 10 Hz, where 100% corresponds to the constant full-crying volume of the (50, 100] stress band.

**SR-CR-2**: S2 shall track changes in crying volume within 1 second after the change (the model drops the volume instantly below 50% stress).

**SR-CR-3**: S2 shall attach a trend flag (rising / stable / falling) to the cry level.

**SR-CR-4**: S2 shall use the fixed microphone provided on the cradle (the microphone setup may not be moved or modified) and its signal conditioning shall be built on the S2 sub-module board.

**SR-CR-5** (display): S2's LCD shall show the current cry level in %, the trend flag, and a live value of the conditioned microphone signal, refreshed at 10 Hz.

**Set of requirements for the Submodule "Decision Making"** **SR-DM-1**: S3 shall estimate the baby's stress level S in 10-100% from the two input signals only (BPM from S1, cry level from S2), with a refresh of at least 1 Hz.

**SR-DM-2**: S3 shall maintain the state of the inverse model (current cell, current K value) and select the next motion (F, A) from the adjacent cells according to the unique decreasing-K path; it shall start at K9 (F5, A5).

**SR-DM-3**: S3 shall detect panic behavior (stress rising despite a "softer" motion = panic block candidate; an immediate stress jump = panic jump) and, on detection, step back to the previous safe motion and retry a different neighbour.

**SR-DM-4**: S3 shall hold every commanded motion for a minimum of 5 seconds before changing, to allow the model to converge (the model may take more than one step to converge to Sopt).

**SR-DM-5**: S3 shall stop all rocking decisions (freeze last safe motion) when the stress estimate reaches 10% (baby calm) and hold it there.

**SR-DM-6** (display): S3's LCD shall show the stress estimate, the current matrix cell (Fi, Aj) and K value, and the decision history of the last 5 steps, refreshed at 1 Hz.

**Set of requirements for the Submodule "Motors drivers"** **SR-MT-1**: S4 shall generate two PWM signals, PWM_F and PWM_A, at a constant 1 kHz carrier frequency and 12 V amplitude, capable of supplying at least 0.8 A.

**SR-MT-2**: S4 shall map the received commands (F1-F5, A1-A5) to the middle of their duty windows (5%, 20%, 40%, 60%, 80%) and apply them with a settling time below 200 ms.

**SR-MT-3**: S4 shall clamp any commanded duty to 90% maximum; if a clamp event occurs it shall flag it on its LCD and send it over the backbone, and it shall never output a duty above 90%.

**SR-MT-4**: S4 shall power down both PWM outputs (duty 0) on command from S3 (emergency / reset), within 100 ms.

**SR-MT-5** (display): S4's LCD shall show the commanded F and A steps, the actual duty values, and the clamp flag, refreshed at 10 Hz.

**SR-MT-6** (watchdog): if S4 receives no SIG_CMD or SIG_ST from S3 for 2 seconds, it shall ease the cradle back to a fixed gentle motion (F1, A1) and flag the lost connection on its LCD.

| **Description 1.2.3** | **Max Points** |
|---|---|
| Distribution of the score : 1,25 point per correct Set of requirements for a Submodule.<br>- Heartbeat<br>- Crying<br>- Decision making<br>- Motor drivers | 5 |

### 1.3 Signals description at submodule level (Sub-system level)

**1.3.1 Internal Signal SIG_HR — heartbeat measurement (S1 → S3)** Heartbeat measurement from S1: current BPM value plus trend flag (rising / stable / falling).

Why: S3 needs a stress-linked input directly. The trend flag also lets S3 distinguish a true stress change from the model's fixed heartbeat delay.

**1.3.2 Internal Signal SIG_CR — crying measurement (S2 → S3)** Crying measurement from S2: normalized cry level in % plus trend flag.

Why: The cry level is the second stress-linked input. We also send a trend flag here. Normalizing at the source (S2) instead of sending raw voltage to S3 removes the need for a second processing stage on S3.

**1.3.3 Internal Signal SIG_CMD — rocking motion command (S3 → S4)** Rocking motion command from S3: a pair (F, A) plus a sequence number.

Why: S3 decides, S4 executes; separating decision from execution is what the V-model asks for [1], and it keeps S4 a simple converter. The sequence number lets S4 detect lost or duplicated commands on the line.

**1.3.4 Internal Signal SIG_ST — decision state (S3, broadcast to every board)** SIG_ST comes from S3 and goes to all boards, once per second. It carries the stress estimate and the (F, A) decision that S3 is running on right now. Its main job is the watchdog in SR-MT-6: if S4 does not get a SIG_ST (or a SIG_CMD) for 2 seconds, it knows S3 is down and eases the cradle back to a fixed gentle motion.

**1.3.5 Internal Signal SIG_STATUS — board health (every board, broadcast)** Every board sends SIG_STATUS once per second. It is a small frame that only says this board is alive. Its purpose is to make a dead board or a broken time-slot rule visible within one second. For S4 it also supports the watchdog in SR-MT-6: no SIG_ST and no SIG_STATUS from S3 means S3 is down, and the cradle eases back to the fixed gentle motion. Together, the five signals close the control loop of Section 1.1.1: two sensed inputs reach S3, one command leaves S3, and two broadcast signals keep every board safe.

| **Description 1.3.1** | **Max Points** |
|---|---|
| Distribution of the score (total 10 points): each signal (description and justification) is scored pro-rata (proportionally, and with a minimum of 3 internal signals considered)<br>• Signal 1<br>• Signal 2<br>• Signal 3<br>• Signal n | 10 |

Table 1: Scoring section 1

| **Final Overview Section 1** | **Max Points** | **Score** |
|---|---|---|
| **Overview Section 1.1** | 10 | |
| **Overview Section 1.2** | 10 | |
| **Overview Section 1.3** | 10 | |
| **Total** | 30 | |

## 2 Design description of the "Communication" protocol

***This submodule is addressed separately in this document because of its central role in the product.***

### 2.1 Goal(s) of the "Communication" protocol

The four boards exchange five internal signals: SIG_HR and SIG_CR go to S3, SIG_CMD goes from S3 to S4, SIG_ST goes from S3 to every board, and SIG_STATUS (board health) goes from every board. The requirements for the protocol, following from the architecture of Section 1, are:

**COMM-1**: the protocol transfers all five signals between the four boards.

**COMM-2**: the updates keep up with the refresh rates of Section 1.

**COMM-3**: a corrupted frame is detected and dropped, and a lost frame is detectable by the receiver.

**COMM-4**: every board shows on its LCD the values of the signals it receives, refreshed at least as often as the signals change.

These four requirements follow directly from the product goal in Section 1.1.1: the protocol is the link that carries the two sensed inputs to the decision maker and the decision to the cradle.

| **Description 2.1** | **Max Points** |
|---|---|
| - The section is short and concisely written. The section considers:<br>a. Goals and requirements of the previously proposed architecture (for the communication between the PYNQ boards) – 1point.<br>b. It includes requirements for the Graphical User Display – 1point. | 2 |

### 2.2 Two Design options for the "Communication" protocol

The hardware already handles the physical layer for us. The UART peripheral of the PYNQ board takes care of the start bit, the stop bit and the baud rate through the course library [3], so we do not design a bit-level layout. What we do design is what the boards put on the wire: a destination and source address, a signal type, a sequence number, a short payload, and a CRC. A gap in the sequence numbers means a frame was lost, a bad CRC means it is dropped. In Option 2 the destination address comes from the I2C hardware itself, because every slave has its own address; the signal type, sequence number, payload and CRC are the same in both options.

**2.2.1 Design Option *1* Topology.** The backbone gives every slot three UART terminals: one send (TX), one receive (RX), and ground. We use these terminals as two shared signal lines: an uplink and a downlink. S1, S2 and S4 connect their send pin AR1 to the uplink, and the decision board S3 connects its receive pin AR0 to the same wire. S3 connects its send pin AR1 to the downlink, and S1, S2 and S4 connect their receive pin AR0 to it. The hub is therefore wired the other way round: it receives on AR0 and sends on AR1, while the other three boards do the opposite. The ground terminal of all four boards is joined. UART itself is a point-to-point, asynchronous link: it has no addressing and no master, so the addressing and the turn-taking are built by us on top of it. The boards run at 9600 baud (8 data bits, no parity, 1 stop bit), the rate set in the UART peripheral of the course bitstream [3]. A board drives its send pin only while it is sending a frame, about 15 ms, and leaves it floating (input) the rest of the time. This is what lets three boards share the uplink: an idle board does not hold the line. Between transmissions the signal lines float, so each line can be pulled to 3.3 V with a 2.2 kΩ resistor for a clean idle level; the link works without the pull-ups, but they help when a wire is long. One of the four boards has a broken AR0 input: the signal arrives at the header pin but never reaches the chip. That board receives on AR2 instead, chosen by a command-line flag. This was measured on the bench, and the demo program supports the AR2 receive pin on any board.

**How it works.** Every board has an address (S1 to S4), and a board keeps a frame when the destination is its own address or the broadcast address. Only one board drives a line at a time. Each board sends in its own

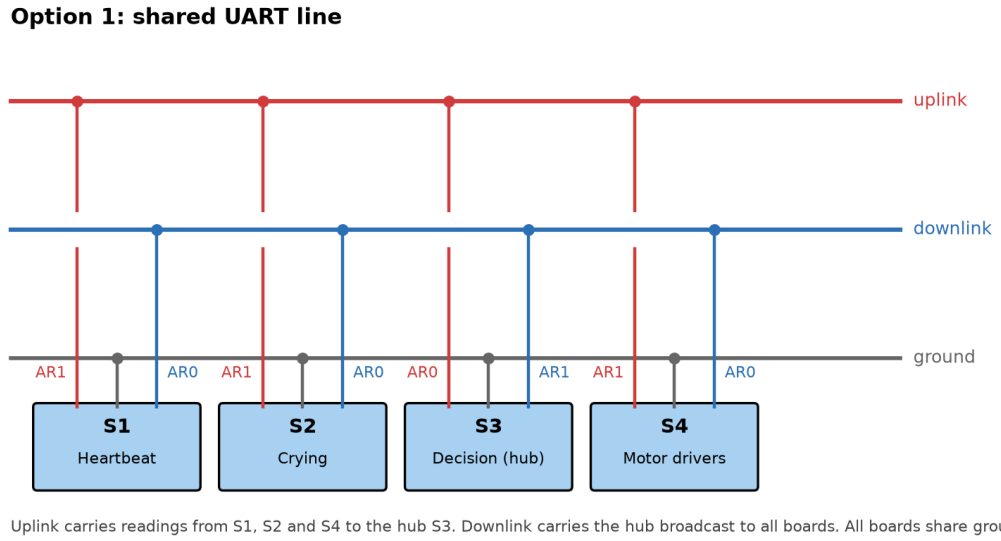



Figure 3: Design Option 1: shared UART line. The uplink joins the send pins of S1, S2 and S4 to the receive pin of the hub S3. The downlink joins the send pin of S3 to the receive pins of the other three boards. All boards share ground.

quarter of every second (S1 at 0 ms, S2 at 250 ms, S3 at 500 ms, S4 at 750 ms), inside a 220 ms window, so two boards never transmit at the same time. The slots are anchored to the frames of S3, which every board receives, so the timing lines itself up and no board clock has to be set. The traffic pattern follows the roles: S1 and S2 send their readings up to S3, S3 broadcasts the decision state down to all boards, and S4 broadcasts its status. Every board sends once per second, so every board can see that the others keep their slots.

**What we build.** A small sender and receiver: the sender packs the frame and puts it on the wire, the receiver waits for the start of a frame, collects the bytes, and checks them before handing the value to the sub-module code. The code is a few hundred lines of C per board, and the board sleeps between frames, so the communication uses almost no CPU.

**Pros.** The option fits the backbone as it is: any board can be placed in any slot, because the wiring pattern is the same for every board. Every board transmits on its own, without being asked, so S4 announces a clamp in its next slot, within one second. Moreover, there is no master to fail: if a sensor board drops out, the other boards keep exchanging frames.

**Cons.** The line is slow compared to Option 2 (9600 baud), but our traffic uses about 4% of it, so the resulting latency is negligible in practice. The hub slot is wired the other way round, so the role of a board must match its wiring: a board started with the wrong role sends into the wrong line. And one board needs the AR2 receive fallback.

#### 2.2.2 Design Option *2*

**Topology.** S3 communicates with the other three boards over two shared wires: one data line (SDA) and one clock line (SCL). S3 is the master, S1, S2 and S4 are slaves, each with its own address. The link runs at 400 kHz, eight times faster than the UART option, and carries the same frame.

**How it works.** S3 initiates every transfer: it addresses a slave by name, and the slave answers. When S3 asks S1 for the BPM, S1 adds its own status to the same answer, so the request and the report happen at once. A slave cannot transmit on its own, which means two boards can never transmit at the same time and nothing collides. The frame checks remain: the link is faster but still error-prone.

**What we build.** On S3, a loop that asks each slave in turn (S1, S2, S4, repeat). On every other board, a small routine that answers when called. The course library already has the I2C driver in both master and

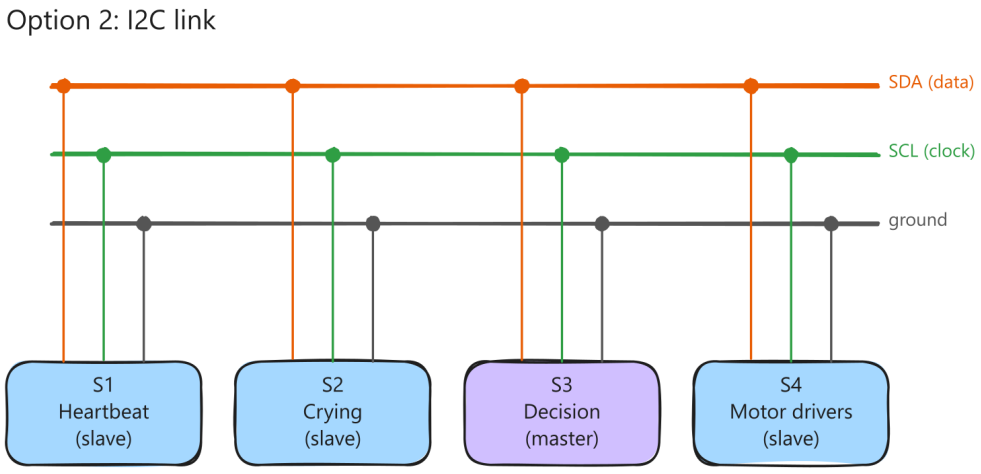



Figure 4: Design Option 2: I2C link, S3 as master, S1/S2/S4 as slaves.

slave mode, so the code is about the same size as Option 1.

**Pros.** The link offers eight times the bandwidth of Option 1. There are no time slots and no collisions: S3 decides which board transmits. Poll and report happen in a single exchange. In addition, the I2C driver is already available in the course library [3].

**Cons.** S3 must do the extra work of master, which adds load to the board that is also doing the decision making. A slave cannot initiate a transfer, so S4 reports a clamp only when its turn comes up, not the moment it happens. Finally, S3 has a different role from the other three boards, so the four slots are no longer identical.

| **Description 2.2** | **Max Points** |
|---|---|
| - The description of two proposed alternative design should contain (5 point per alternative design choice).:<br>a. a high level drawing (block diagram) with clear inputs/outputs (1 point),<br>b. a protocol description (1 point),<br>c. I/O details (1 point),<br>d. key programming elements (1point),<br>e. a list of Pro and/ or Cons (1 point).<br>- Points for Design option 1: (max 5 points)<br>- Points for Design option 2: (max 5 points) | 10 |

### 2.3 Final Design choice for the "Communication" protocol

We choose Option 1, the shared UART line. The table compares the two options on the parameters that decide the choice.

| Parameter | Option 1: shared UART line | Option 2: I2C link (S3 master) |
|---|---|---|
| How it is wired | two shared signal lines (uplink and downlink) plus ground; the hub receives on the uplink and sends on the downlink; any board can take the hub slot | two shared wires (SDA and SCL) plus ground, S3 as master |
| How fast | 9600 baud, sufficient for our traffic (about 4% used) | 400 kHz, eight times faster |
| Who transmits when | the boards take turns in fixed time slots | S3 calls each slave in turn |
| S4 reports a clamp | on its own, in its next slot (within one second) | only when its turn comes up |
| Same job on every board | yes, one program for all four boards; the role is a number on the command line | no, S3 is the master |
| How much we write | a sender, a receiver, a few hundred lines of C | about the same, the I2C driver is ready in the course library |

Three reasons. First, and most important, an S4 clamp has to be reported without waiting to be asked, and only Option 1 lets S4 transmit on its own. Second, the backbone is identical in every position, and Option 1 is the only option that keeps all four boards doing the same job, so any board can go in any slot. Third, Option 1 adds nothing to S3, which is already the busiest board. The slower line is acceptable because our traffic uses under 2% of it. The choice remains open: if the time-slot rule proves hard to keep in practice, the I2C option is a reliable fallback.

| Description 2.3 | Max Points |
|---|---|
| - A comparison of proposed design (alternatives and chosen one) is given with respect to relevant design parameters – 2,5 point.<br>- A reason for the final design choice is given – 2,5 point. | 3 |

### 2.4 Tests of the "Communication" protocol

**First test (Demo 1) [5].** All four boards are connected to the backbone, each running only the communication code and a simulated signal source: S1 slowly moves the BPM up and down between 60 and 240, S2 steps the cry level up and down, S3 echoes the stress estimate, and S4 echoes the duty it was told to. This test checks the whole chain with real boards. Pass: every LCD shows the value that was sent, the values change during the run, and the scope shows clean frames for ten minutes with no corruption.

**Second test (integration, before Demo 2).** Now the real sub-modules run, and the four requirements are checked one by one. For COMM-1, we record for every signal who sends it, who receives it, and how often it updates, and we compare these records with Section 1. For COMM-2, we measure with the scope on S4 the time from a SIG_CMD change to the resulting PWM change, which must stay under 200 ms. For COMM-3, a small lab harness corrupts frames on purpose through bit flips and drops; we check that no bad frame is accepted, that every lost frame is detected, and that the PWM duty stays within the allowed values. For COMM-4, we check on every board that its LCD shows the values of the signals it is supposed to show.

Once the hardware is available, the measured values go into Section 4.

| Description 2.4 | Max Points |
|---|---|
| • The proposed test is well described and effective at verifying the given specifications - 2 points<br>• The test results are given with full details (with all specifications) – 1,5 points<br>• The Justification and explanation are correct -1,5 points | 10 |

### 2.5 Justification to fall back to the back-up solution provided

We keep Option 2 as the back-up. If the shared UART line proves unreliable on the real backbone, for example if frames keep getting corrupted in the ten-minute test, we switch to it. The switch is deliberately inexpensive: both options use the same frame, the same checks, and the same sender and receiver code. Only the wiring and the driver change. The five signals stay the same, so the LCDs and the safety checks keep working as before. Because Option 2 is fully described in Section 2.2, choosing Option 1 now is low-risk: the plan B is not a guess, it is a working design.

Table 2: Scoring Section 2

| Overview Section 2 | Max Points | Score |
|---|---|---|
| **2.1** | 2 | |
| **2.2** | 10 | |
| **2.3** | 3 | |
| **2.4** | 10 | |
| **Total** | 25 | |

## Appendix B: Referenced Documents

This document follows the official Rock Your Baby design document template and is written against the following documents.

| Ref. | Document Name | Issue Date |
|---|---|---|
| [1] | RYB Main Document (1_RYB_Main_Document_Final_2627.pdf), Canvas 5EWC0, module 106032, file 7246072 | 2026 |
| [2] | RYB System Overview 2026-2027 (9_RYB_System_Overview_2627.pdf), Canvas 5EWC0, module 106032, file 7246092 | 2026 |
| [3] | libpynq, the RYB course library (source: Team 34 repository, src/libpynq) | 2026 |
| [4] | RYB Setup Start-up Guide (RYB_Setup_Start_up_Guide.pdf), Canvas 5EWC0, module 106032 | 2026 |
| [5] | Rubric Demos (2627_RYB_Rubric_Demos.pdf), Canvas 5EWC0, module 106032, file 7250230 | 2026 |
| [6] | Dynamisches Mikrofon DM 250 datasheet (3_300068-da-01-en-Dynamisches_Mikrofon_DM_250-1.pdf), Canvas 5EWC0, module 106032 | 2026 |
| [7] | Design Coach assignment (2627_Design_Coach_Team_Assignment.xlsx), Canvas 5EWC0, module 106032, file 7260221 | 2026 |